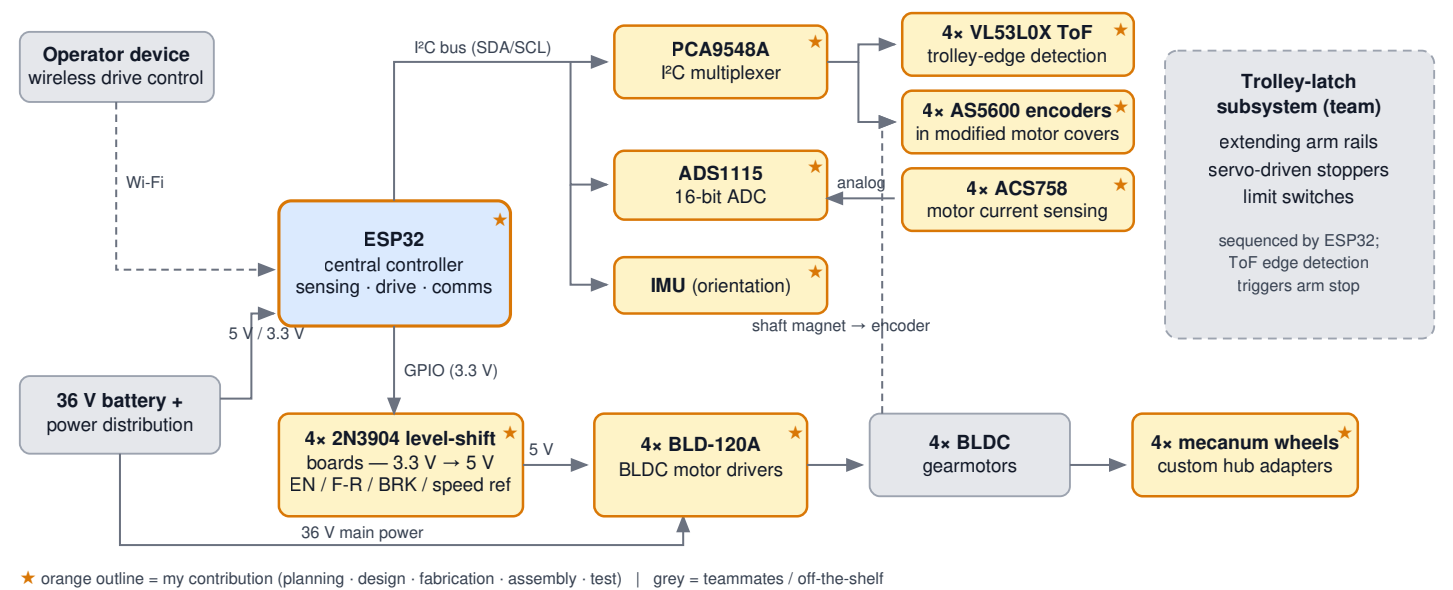




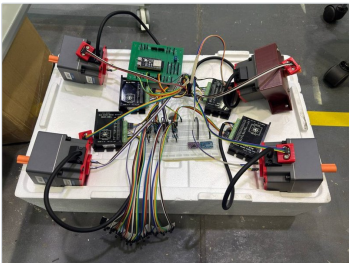
★ orange outline = my contribution (planning · design · fabrication · assembly · test) | grey = teammates / off-the-shelf

MY CONTRIBUTIONS

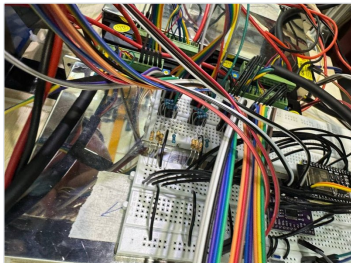
- 1. Planned & assembled the base electrical system (19 components).**
Architected the full sensing/drive circuit around a single ESP32: 4× VL53L0X ToF rangefinders and 4× AS5600 encoders behind a PCA9548A I²C multiplexer (resolves fixed-address clashes on the shared bus), 4× ACS758 current sensors digitised by an ADS1115 ADC on the same I²C bus, an IMU, and 4× BLD-120A BLDC drivers.
- 2. Designed the 3.3 V→5 V motor-driver interface.** ESP32 logic levels were not reliably registered by the BLD-120A's 5 V control inputs — the hardest debugging problem of the build. Solved it with a 2N3904 interface board per driver: each GPIO drives an NPN via a 10 kΩ base resistor (100 kΩ pull-down for a defined off-state), and the transistor sinks the driver's internally

- pulled-up control line (EN, F/R, BRK) to COM; 1.5 kΩ sets the speed reference. Schematic and PCB drawn in KiCad.
- 3. Wired and soldered** most of the sensor harnesses and connections across the base circuit.
 - 4. Designed & fabricated a modified motor cover** housing each AS5600 encoder PCB, with the diametric magnet aligned on the motor shaft axis.
 - 5. Designed & fabricated wheel-hub adapters and motor mounts** coupling the mecanum wheels to the gearmotor shafts, plus the ToF sensor mounts.
 - 6. Ran the integration drive test:** drove the fully constructed base untethered on 36 V battery power under ESP32 wireless control.

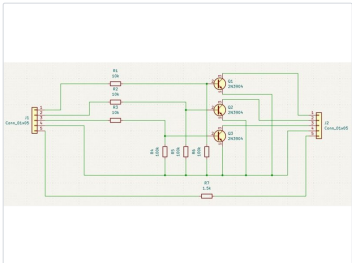
SUPPORTING FIGURES



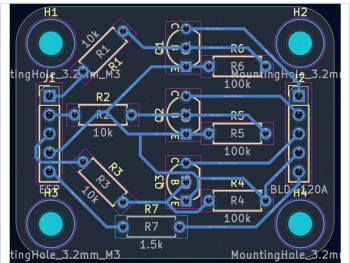
Base electronics, assembled (pre-mounting)



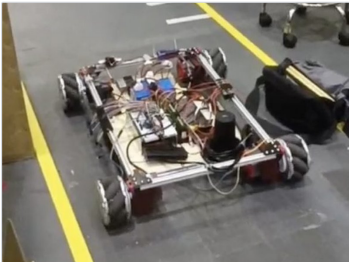
Level-shift circuit used in the drive test



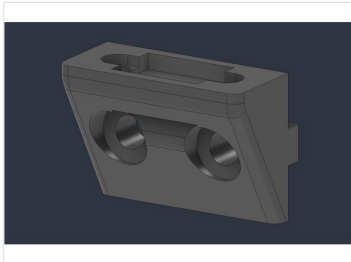
Level-shifter schematic (one motor)



Level-shifter PCB layout (one motor)



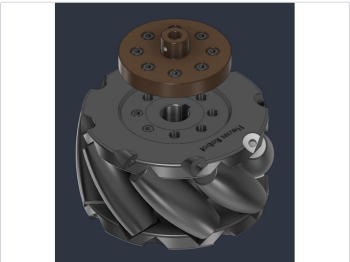
Untethered drive test (36 V battery)



ToF sensor mount (CAD)



Modified motor cover + AS5600 (exploded CAD)



Wheel-hub adapter, mecanum wheel (CAD)